

Programmatic navigation in React Router allows developers to redirect users based on actions like form submissions or authentication, using hooks like `useNavigate` (v6) or `useHistory` (v5)—with `useNavigate` being the modern replacement that offers simpler, function-based navigation.

The `useParams` hook extracts dynamic route parameters (e.g., `:id`) from the URL, enabling components to render content based on those values—such as fetching and displaying a specific user's details on a profile page.